

★ WEEK 12 · PART 3 · PRACTICE

Discretize and Calibrate, Worked by Hand

Step an SDE, Fit It, and Price by Simulation

Euler-Maruyama, Milstein, and a calibration workflow

Part 3's numerical toolkit, with the arithmetic shown. We take an Euler step by hand and see the negative-rate pitfall Milstein repairs, then calibrate Vasicek from history and price a bond by simulation - checking it against the closed form.

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PART 3 · APPLICATIONS

From Formula to Simulation

Case A: an Euler-Maruyama step, the negative-rate pitfall it can hit, and the Milstein correction that repairs it. Case B: calibrate Vasicek via the AR(1) bridge and price a bond by simulation against the closed form. Then a Claude Code lab.

What Part 3 Told Us

Now we step and fit - with numbers

1

Euler-Maruyama steps an SDE

Drift scaled by dt , noise by $\sqrt{dt}$. Simple and general - but it makes discretization error and can break positivity.

2

Milstein adds one correction

For a level-dependent diffusion, an extra term sharpens accuracy and helps keep rates positive.

3

Calibrate, then simulate to price

Fit the parameters from history or the curve, then simulate, discount, and average to price - checked against the closed form.

Everything here is a calculator exercise: one Euler step, one calibration, one simulated price.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from an SDE or a data series.
- Take a step, calibrate, or simulate by hand.
- Then read the numerical lesson.

What you should watch

- How a big shock can push Euler below zero.
- How Milstein repairs it.
- How the simulated price matches the formula.

An Euler Step, and Its Pitfall

Where discretization breaks positivity

To simulate a rate model, we replace the continuous SDE with discrete steps. Euler-Maruyama is the simplest: move by the drift times dt and the noise times $\sqrt{dt}$. We take one step of CIR by hand, then see how a large negative shock can push the discretized rate below zero - and how Milstein repairs it

The setup

- CIR: $\kappa = 0.30$, $\theta = 5\%$, $\sigma = 0.20$.
- Monthly step, starting rate 1%.
- Near the floor, where trouble lives.

What we'll do by hand

- Compute the drift and noise pieces.
- Step with a normal and a large shock.
- Apply the Milstein correction.

The Euler-Maruyama Step

Drift by dt, noise by root-dt

$$r_{t+\Delta t} = r_t + \underbrace{\kappa(\theta - r_t)\Delta t}_{\text{drift} \times \Delta t} + \underbrace{\sigma\sqrt{r_t}\sqrt{\Delta t}z}_{\text{noise} \times \sqrt{\Delta t}} \quad (z \sim N(0, 1))$$

The drift piece

- The reversion, scaled by the time step.
- It's deterministic and usually small.
- It nudges the rate toward theta.

The noise piece

- The diffusion, scaled by root-dt.
- A random normal shock each step.
- For CIR it also scales with root-r.

One Euler Step

The drift and noise pieces

$$\begin{aligned}\text{drift} &= 0.30(0.05 - 0.01)\left(\frac{1}{12}\right) = 0.0010 \\ \text{noise coef} &= 0.20\sqrt{0.01}\sqrt{\frac{1}{12}} = 0.00577\end{aligned}$$

At a 1% rate, the drift adds a tiny 0.0010, while the noise coefficient is 0.00577 per unit shock - small, but a large multiple of it can dominate the step.

The Pitfall, and the Fix

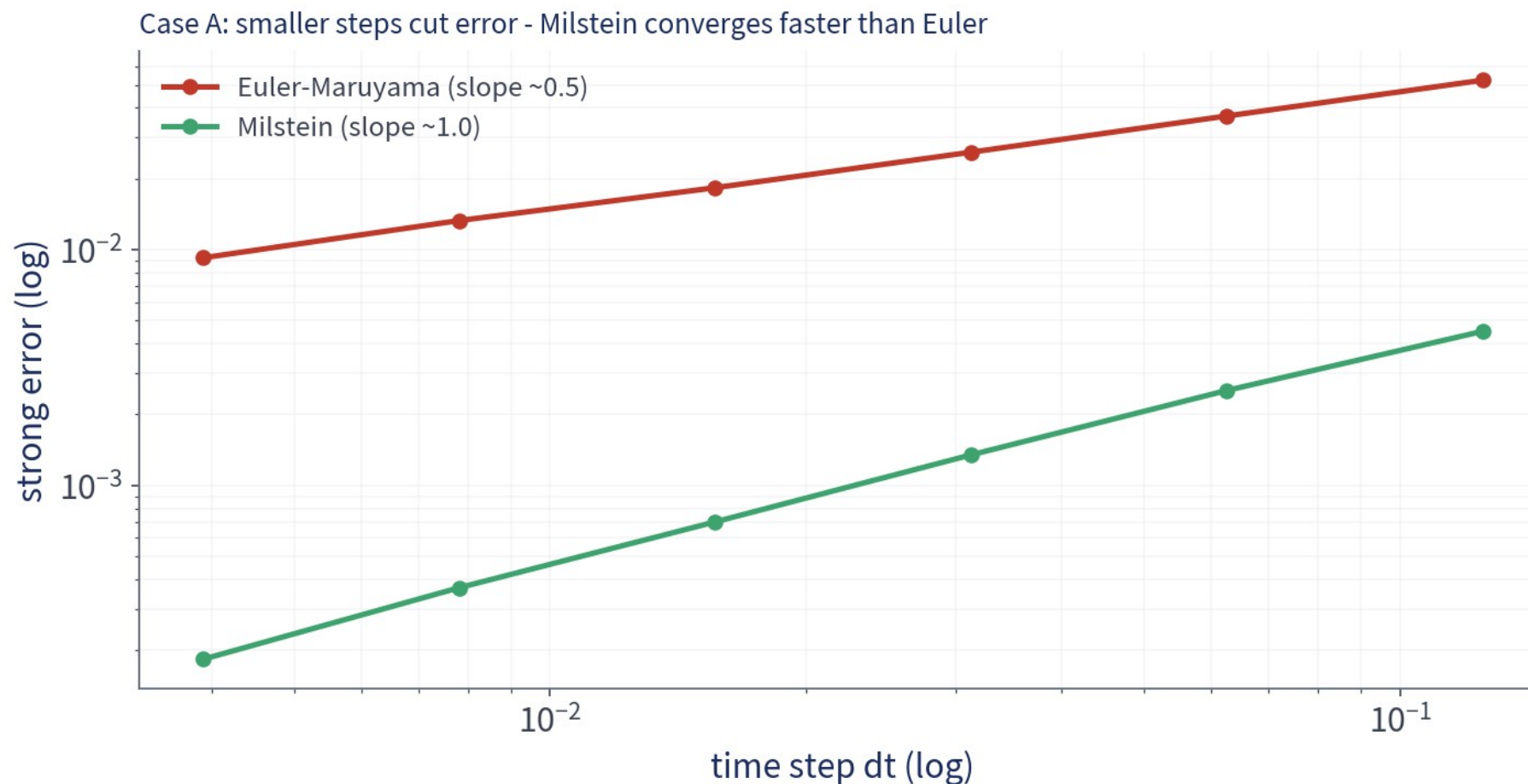
A big shock breaks the floor

$$z = -3 : r_{t+\Delta t} = 0.01 + 0.0010 - 3(0.00577) = -\mathbf{0.0063} \text{ (negative!)}$$
$$\text{Milstein} + \frac{\sigma^2}{4}(z^2 - 1)\Delta t = +0.0067 \Rightarrow r_{t+\Delta t} = +0.0004 \text{ (repaired)}$$

A minus-three shock drives the Euler step to -0.0063 - a negative rate that continuous CIR can never reach. The Milstein correction adds 0.0067 and repairs it back above zero.

Smaller Steps, Less Error

On a log-log plot, both schemes' error shrinks as the step size falls - but Milstein falls twice as fast (slope near 1) as Euler (slope near 0.5). For the same accuracy, Milstein lets you take far larger, cheaper steps.



Choose the Scheme Deliberately

Accuracy and positivity both matter

Euler's weaknesses

- Error shrinks only with root-dt.
- A big shock can push rates negative.
- Fine for rough work, risky near a floor.

When to reach for Milstein

- Level-dependent diffusion, like CIR's root-r.
- Faster convergence, better positivity.
- One extra term, well worth it near zero.

Calibrate, Then Price by Simulation

Fit the model, then trust it

A model is only useful once it's fit to data. We calibrate Vasicek from a monthly rate history through the AR(1) bridge - one more time - recovering the reversion speed, long-run level, and volatility. Then we price a five-year bond by simulation and check it against the affine closed form from Part 2

The history

- Monthly AR(1): coefficient 0.96.
- Intercept 0.002, residual sd 0.005.
- Sampled monthly - dt is one twelfth.

What we'll do by hand

- Map to kappa, theta, and sigma.
- Price a 5-year bond by simulation.
- Compare to the closed form.

Calibration via the AR(1) Bridge

Three parameters from a regression

$$\kappa = -\frac{\ln \phi}{\Delta t}, \quad \theta = \frac{c}{1-\phi}, \quad \sigma = \sigma_\varepsilon \sqrt{\frac{2\kappa}{1-\phi^2}} \quad (\text{AR}(1) \rightarrow \text{Vasicek})$$

The mapping

- kappa from the coefficient, as in Week 11.
- theta from the intercept over one-minus-phi.
- sigma from the residual sd, annualized.

The principle

- Estimate discretely with a simple regression.
- Interpret the result in continuous time.
- One line of code, three model parameters.

The Fitted Parameters

kappa, theta, sigma from history

$$\begin{aligned}\kappa &= -\ln(0.96) \times 12 = \mathbf{0.49}/\text{yr} \quad (\text{half-life } 1.4 \text{ yr}) \\ \theta &= \frac{0.002}{0.04} = \mathbf{5.0\%}, \quad \sigma = 0.005\sqrt{12.5} = \mathbf{1.77\%}\end{aligned}$$

The AR(1) fit gives a reversion speed of 0.49 per year - a 1.4-year half-life - a long-run level of 5%, and a volatility of 1.77%. A full Vasicek model, from one regression.

Price by Simulation

Simulate, discount, average

$$P = \mathbb{E} \left[e^{-\int_0^T r dt} \right] : \quad \text{Monte Carlo } 0.8099 \pm 0.0002 \approx \text{closed form } 0.8096$$

The workflow

- Simulate many short-rate paths with Euler.
- Discount each by its own integrated rate.
- Average the discount factors.

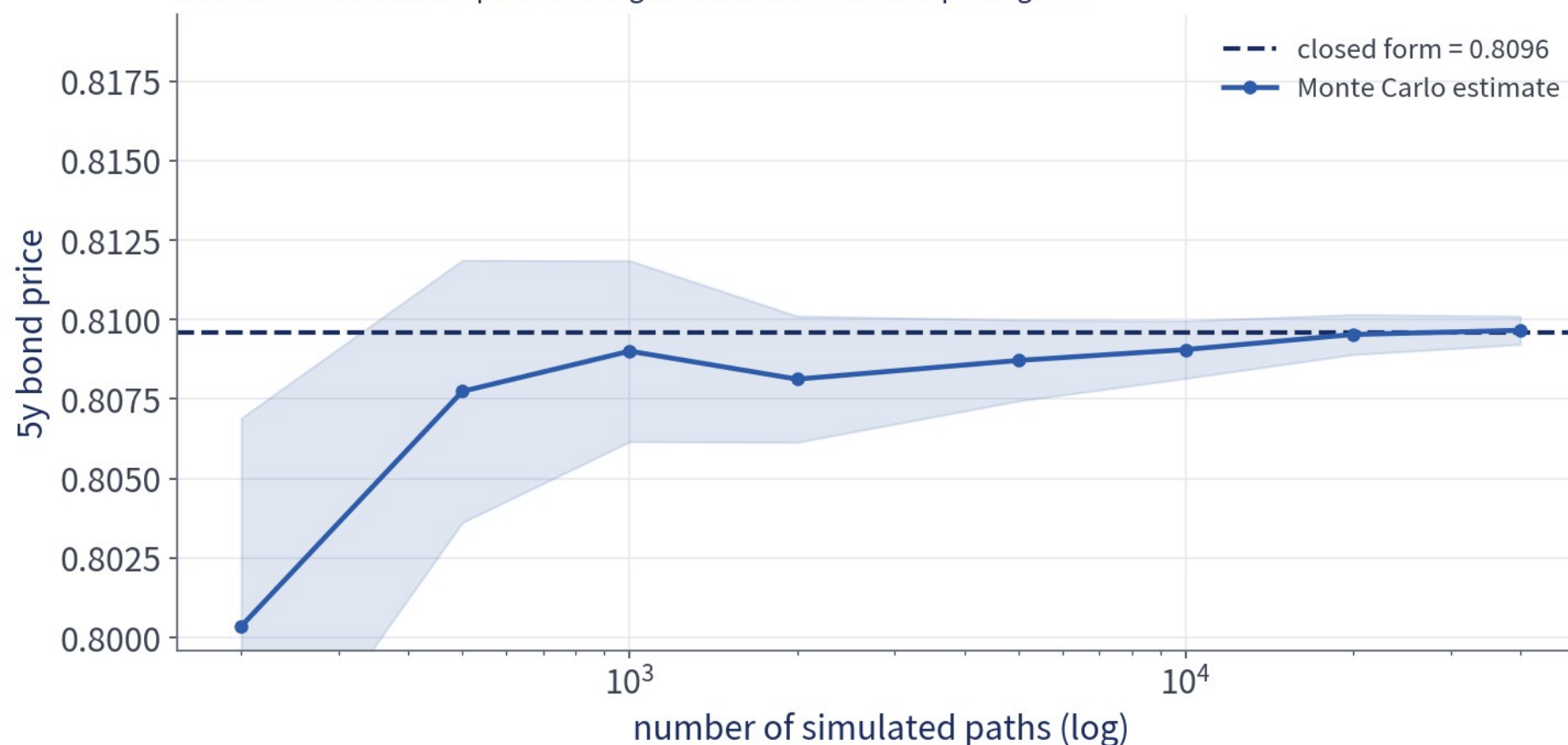
The check

- The Monte Carlo price is 0.8099.
- The closed form is 0.8096.
- They agree - the simulation is validated.

Simulation Meets the Formula

With only a few hundred paths the Monte Carlo price is noisy, but as the path count grows it converges tightly onto the closed-form value. The shrinking band is the shrinking standard error - and its landing on the dashed line confirms the simulation is correct

Case B: the Monte Carlo price converges to the closed form as paths grow



Why Both Data and Simulation

Fit to reality, then compute anything

Two sources of data

- History calibrates the real-world dynamics.
- Prices calibrate to today's observed curve.
- Which you use depends on the question.

Why simulate at all

- The closed form only exists for simple models.
- Simulation prices anything - once validated.
- Check against the formula where you can.

The Full Numerical Workflow

Discretize, calibrate, simulate, g
Part 3 turns the theory into computation. Discretize the SDE to step it - choosing Euler or Milstein for the accuracy and positivity you need. Calibrate the parameters from history or prices. Then simulate, discount, and average to price - validating against the closed form wherever one exists. That loop is how rate models

Discretize (Case A)

- Euler for simplicity, Milstein for accuracy.
- Watch the negative-rate pitfall.
- Smaller steps, less error.

Calibrate and price (Case B)

- AR(1) bridge recovers the parameters.
- Simulate, discount, average to price.
- Validate against the closed form.

Reproduce the Step and the Price

Direct Claude Code to check your arithmetic

GOAL Verify the Euler step and Milstein fix, calibrate Vasicek, and price by simulation.

[INPUT] Prompt · Claude Code

Write and run part3_practice.py that:

CASE A - Euler:

- * for CIR ($k=0.30$, $\theta=0.05$, $\sigma=0.20$, $dt=1/12$, $r=0.01$), print the Euler step for $z=+1$ and $z=-3$;
- * add the Milstein term and show it repairs $z=-3$.

CASE B - calibrate + price:

- * from AR(1) $\phi=0.96$, $c=0.002$, $se=0.005$, print κ , θ , σ ;
- * price a 5y bond by simulation and compare to the closed form.

Match every printed number to the slides.

Check against the slides

- Euler $z=-3$: -0.0063 (negative).
- Milstein repairs to +0.0004.
- κ 0.49, θ 5%, σ 1.77%.
- MC 0.8099 vs closed 0.8096.

Deliverable: part3_practice.py whose printed numbers match the Euler step, calibration, and price.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does the noise scale in an Euler step?

With root-dt, not dt

Q2 Why can Euler give a negative rate?

**A big shock
overshoots the zero
floor**

Q3 How do you price a bond by simulation?

**Simulate, discount
each path, average**

All three come from the Euler step, the negative-rate pitfall, and the simulation workflow.

Three Pitfalls to Avoid

Where discretization trips people

Common mistakes

- "Scale the noise by dt like the drift."
- "Euler is always accurate enough."
- "Trust a simulated price with no check."

The reality

- Noise scales with $\sqrt{dt}$ - it dominates small steps.
- Near a floor, Euler breaks positivity; use Milstein.
- Validate against the closed form whenever one exists.

Two Calculations to Remember

Part 3 applications in one screen

1

Step carefully, and pick the scheme

An Euler step of CIR at 1% had a drift of 0.0010 and a noise coefficient of 0.00577; a minus-three shock pushed it to -0.0063, a negative rate, which the Milstein correction of 0.0067 repaired. Smaller steps and Milstein cut the error.

2

Calibrate, then price by simulation

The AR(1) bridge recovered $\kappa = 0.49$, $\theta = 5\%$, and $\sigma = 1.77\%$ from a monthly history. Simulating, discounting, and averaging gave a 5-year price of 0.8099, matching the closed form's 0.8096 - the simulation validated.

3

The full numerical workflow

Discretize to step, calibrate to fit, simulate to price, and check against the formula. That loop takes a rate model from the page to a number you can trade on.

Next — Week 13: the course turns from continuous-time models toward machine learning for financial time series.